



AI reinvention made real

Accenture Innovation Challenge 2026

Team details

TEAM NAME: AutoSynch



Sunand Chitikala

College: National Institute of
Technology Karnataka, Surathkal
Stream: Mechanical Engineering
Year of graduation: 2028



Ajay D Kamath

College: National Institute of
Technology Karnataka, Surathkal
Stream: Mining
Year of graduation:2028

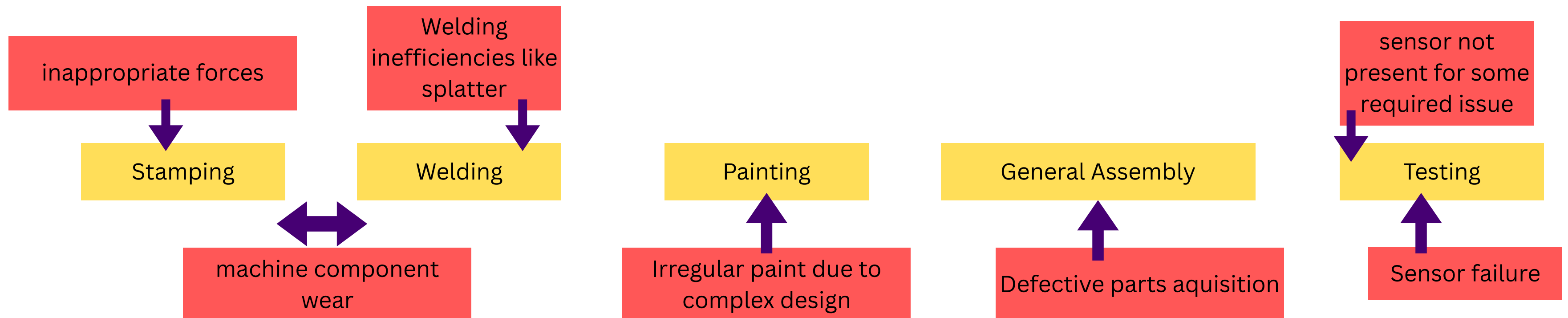
Problem Statement

A vehicle assembly line (VAL) at industry consists of 4 sections which are stamping/welding, painting, general assembly and inspection where only the 3rd process which include adding inner designs and parts like seats, wiring, etc are mostly done with human supervision.

The stamping is done with dies while welding are done through various processes such as spot welding, arc welding, laser welding, etc based on various joints and strength requirement. The welding process does about 5000 joints and the process is highly automated with little or no supervision.

The automation process has sensors which only account for critical mechanical failures in machines such as a part getting stuck, etc while defects cannot be monitored continuously until the testing phase of the process. Plants usually check defects and imperfections till micrometer scales and hence the following issues need to be taken seriously to prevent loss when the line has to discard the body. When a new design enters for production usual challenges faced may include lot of tension/ compression spots during stamping which can plastically deform or fail the structure. Also welding defects of joints, splatter can be primary concerns.

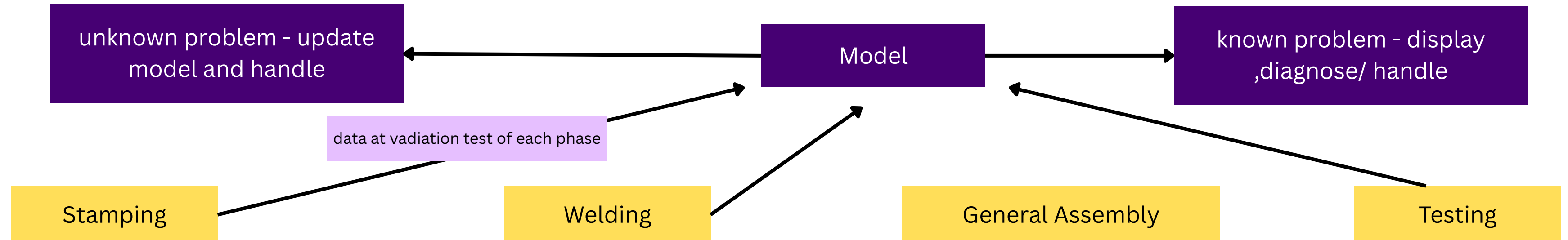
Also, machine wear due to many factors which occur would resemble a pattern of emerging unnoticeable imperfections until a critical failure may occur in VAL or during test phase. Also, components of general assembly are mostly outsourced from many different industries. There can be manufacturing defects in some components which create a lot of confusion and losses.

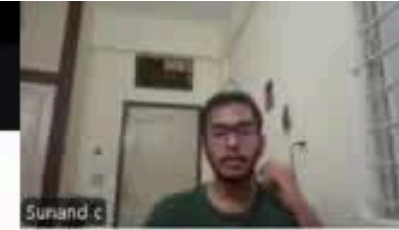


Solution Approach

The model should be trained using various sensor/ materials data at each stage like triangulation sensors, position encoders, infrared thermography, etc based on patterns of data wrt model type, process it has gone through, time series curves, etc and their comparisons with known problems such as tool wear, sensor failure, design issues, manufacturer material issues in general assembly, etc so that the model can classify the issues and also give a live feedback to the technicians to diagnose the problem. Many problems would have already been dealt with during design and research phase but maintenance and supervision of a 95-100% automated stamping, welding and painting requires analysis which might not be significant at first but can cause a critical failure at some point. Also, the model should be able to give out unknown patterns which are not yet known so that we can update accordingly. These solutions would solve many problems because up to 40% of the problems in VAL's are due to sensor/ software problems.

The approach of model building is to make a CNN based decision tree classifier which has proven effective in various areas in identifying types of patterns.





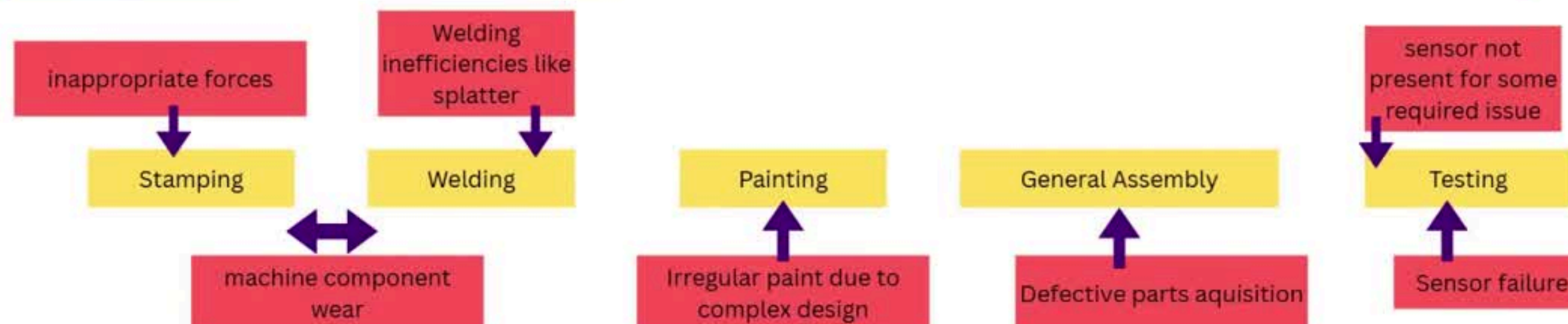
Problem Statement

A vehicle assembly line (VAL) at industry consists of 4 sections which are stamping/welding, painting, general assembly and inspection where only the 3rd process which include adding inner designs and parts like seats, wiring, etc are mostly done with human supervision.

The stamping is done with dies while welding are done through various processes such as spot welding, arc welding, laser welding, etc based on various joints and strength requirement. The welding process does about 5000 joints and the process is highly automated with little or no supervision.

The automation process has sensors which only account for critical mechanical failures in machines such as a part getting stuck, etc while defects cannot be monitored continuously until the testing phase of the process. Plants usually check defects and imperfections till micrometer scales and hence the following issues need to be taken seriously to prevent loss when the line has to discard the body. When a new design enters for production usual challenges faced may include lot of tension/ compression spots during stamping which can plastically deform or fail the structure. Also welding defects of joints, splatter can be primary concerns.

Also, machine wear due to many factors which occur would resemble a pattern of emerging unnoticeable imperfections until a critical failure may occur in VAL or during test phase. Also, components of general assembly are mostly outsourced from many different industries. There can



Thank you

